```cpp
    //The Destructor - CRITICAL for Manual Memory
    ~NodeList() {
        ListNode* current = head;
        while (current != nullptr) {
            ListNode* nextNode = current->next;
            //delete current; // clean up each node
            current = nextNode;
        }
        head = nullptr;
    }
};
```

I commented out the delete current line to create a memory leak.

```
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$ g++ -g manual_list.cpp -o manual_list_leaky
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$ valgrind --leak-check=full ./manual_list_leaky
==62428== Memcheck, a memory error detector
==62428== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==62428== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==62428== Command: ./manual_list_leaky
==62428==
[1: Router-A] -> [2: Switch-B] -> NULL
After removing ID 1:
[2: Switch-B] -> NULL
==62428==
==62428== HEAP SUMMARY:
==62428==     in use at exit: 48 bytes in 1 blocks
==62428==   total heap usage: 4 allocs, 3 frees, 74,848 bytes allocated
==62428==
==62428== 48 bytes in 1 blocks are definitely lost in loss record 1 of 1
==62428==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==62428==    by 0x10A637: NodeList::addNode(int, std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >) (manual_list.cpp:22)
==62428==    by 0x10A47B: main (manual_list.cpp:78)
==62428==
==62428== LEAK SUMMARY:
==62428==    definitely lost: 48 bytes in 1 blocks
==62428==    indirectly lost: 0 bytes in 0 blocks
==62428==      possibly lost: 0 bytes in 0 blocks
==62428==    still reachable: 0 bytes in 0 blocks
==62428==         suppressed: 0 bytes in 0 blocks
==62428==
==62428== For lists of detected and suppressed errors, rerun with: -s
==62428== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$
```

^ the above shows the memory leak found with valgrind

```cpp
int main() {
    NodeList myNodes;
    myNodes.addNode(1, "Router-A");
    myNodes.addNode(2, "Switch-B");
    myNodes.display();

    myNodes.removeNode(1);
    cout << "After removing ID 1: " << endl;
    myNodes.display();

    ListNode* extraNode = new ListNode{99, "Bug", nullptr};
    delete extraNode;
    delete extraNode; // ERROR: Double free!

    return 0;
}
```

^double delete bug

```
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$ g++ -g manual_list.cpp -o manual_list_bug2
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$ valgrind ./manual_list_bug2
==64973== Memcheck, a memory error detector
==64973== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==64973== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==64973== Command: ./manual_list_bug2
==64973==
[1: Router-A] -> [2: Switch-B] -> NULL
After removing ID 1:
[2: Switch-B] -> NULL
==64973== Invalid read of size 8
==64973==    at 0x49C7AE4: std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<cha
r> >::~basic_string() (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==64973==    by 0x10A80B: ListNode::~ListNode() (manual_list.cpp:6)
==64973==    by 0x10A593: main (manual_list.cpp:87)
==64973==  Address 0x4e1d5a8 is 8 bytes inside a block of size 48 free'd
==64973==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_m
emcheck-amd64-linux.so)
==64973==    by 0x10A582: main (manual_list.cpp:86)
==64973==  Block was alloc'd at
==64973==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-am
d64-linux.so)
==64973==    by 0x10A508: main (manual_list.cpp:85)
==64973==
==64973== Invalid free() / delete / delete[] / realloc()
==64973==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_m
emcheck-amd64-linux.so)
==64973==    by 0x10A5A0: main (manual_list.cpp:87)
==64973==  Address 0x4e1d5a0 is 0 bytes inside a block of size 48 free'd
==64973==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_m
emcheck-amd64-linux.so)
==64973==    by 0x10A582: main (manual_list.cpp:86)
==64973==  Block was alloc'd at
==64973==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-am
d64-linux.so)
==64973==    by 0x10A508: main (manual_list.cpp:85)
==64973==
==64973==
==64973== HEAP SUMMARY:
==64973==     in use at exit: 0 bytes in 0 blocks
==64973==   total heap usage: 5 allocs, 6 frees, 74,896 bytes allocated
==64973==
==64973== All heap blocks were freed -- no leaks are possible
==64973==
==64973== For lists of detected and suppressed errors, rerun with: -s
==64973== ERROR SUMMARY: 2 errors from 2 contexts (suppressed: 0 from 0)
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$
```

This is the double delete bug ^

```cpp

int main() {
    NodeList myNodes;
    myNodes.addNode(1, "Router-A");
    myNodes.addNode(2, "Switch-B");
    myNodes.display();

    myNodes.removeNode(1);
    cout << "After removing ID 1: " << endl;
    myNodes.display();

    ListNode* ghostNode = new ListNode{100, "Ghost", nullptr};
    delete ghostNode;
    cout << ghostNode->name << endl; // This is the "Use-After-Free"

    return 0;
}


```

^use after free bug

```
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$ g++ -g manual_list.cpp -o manual_list_bug3
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$ valgrind --leak-check=full ./manual_list_bug3
==66825== Memcheck, a memory error detector
==66825== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==66825== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==66825== Command: ./manual_list_bug3
==66825==
[1: Router-A] -> [2: Switch-B] -> NULL
After removing ID 1:
[2: Switch-B] -> NULL
==66825== Invalid read of size 8
==66825==    at 0x49CAD44: std::basic_ostream<char, std::char_traits<char> >& std::operator<< <char, std::char_traits<char>, std:
asic_string<char, std::char_traits<char>, std::allocator<char> > const&) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==66825==    by 0x10A59C: main (manual_list.cpp:87)
==66825==  Address 0x4e1d5b0 is 16 bytes inside a block of size 48 free'd
==66825==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A582: main (manual_list.cpp:86)
==66825==  Block was alloc'd at
==66825==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A508: main (manual_list.cpp:85)
==66825==
==66825== Invalid read of size 8
==66825==    at 0x49CAD48: std::basic_ostream<char, std::char_traits<char> >& std::operator<< <char, std::char_traits<char>, std:
asic_string<char, std::char_traits<char>, std::allocator<char> > const&) (in /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.33)
==66825==    by 0x10A59C: main (manual_list.cpp:87)
==66825==  Address 0x4e1d5a8 is 8 bytes inside a block of size 48 free'd
==66825==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A582: main (manual_list.cpp:86)
==66825==  Block was alloc'd at
==66825==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A508: main (manual_list.cpp:85)
==66825==
==66825== Invalid read of size 1
==66825==    at 0x4B9EB04: _IO_new_file_xsputn (fileops.c:1219)
==66825==    by 0x4B9EB04: _IO_file_xsputn@@GLIBC_2.2.5 (fileops.c:1197)
==66825==    by 0x4B91A11: fwrite (iofwrite.c:39)
==66825==    by 0x49B5DC3: std::basic_ostream<char, std::char_traits<char> >& std::__ostream_insert<char, std::char_traits<char> >
x86_64-linux-gnu/libstdc++.so.6.0.33)
==66825==    by 0x10A59C: main (manual_list.cpp:87)
==66825==  Address 0x4e1d5bc is 28 bytes inside a block of size 48 free'd
==66825==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A582: main (manual_list.cpp:86)
==66825==  Block was alloc'd at
==66825==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A508: main (manual_list.cpp:85)
```

scroll

```
==66825== Invalid read of size 1
==66825==    at 0x4853980: mempcpy (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x4B9EA3F: _IO_new_file_xsputn (fileops.c:1236)
==66825==    by 0x4B9EA3F: _IO_file_xsputn@@GLIBC_2.2.5 (fileops.c:1197)
==66825==    by 0x4B91A11: fwrite (iofwrite.c:39)
==66825==    by 0x49B5DC3: std::basic_ostream<char, std::char_traits<char> >& std::__ostream_insert<char, std::char_traits<char> >(std::ba
x86_64-linux-gnu/libstdc++.so.6.0.33)
==66825==    by 0x10A59C: main (manual_list.cpp:87)
==66825==  Address 0x4e1d5b8 is 24 bytes inside a block of size 48 free'd
==66825==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A582: main (manual_list.cpp:86)
==66825==  Block was alloc'd at
==66825==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A508: main (manual_list.cpp:85)
==66825==
==66825== Invalid read of size 1
==66825==    at 0x485398E: mempcpy (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x4B9EA3F: _IO_new_file_xsputn (fileops.c:1236)
==66825==    by 0x4B9EA3F: _IO_file_xsputn@@GLIBC_2.2.5 (fileops.c:1197)
==66825==    by 0x4B91A11: fwrite (iofwrite.c:39)
==66825==    by 0x49B5DC3: std::basic_ostream<char, std::char_traits<char> >& std::__ostream_insert<char, std::char_traits<char> >(std::ba
x86_64-linux-gnu/libstdc++.so.6.0.33)
==66825==    by 0x10A59C: main (manual_list.cpp:87)
==66825==  Address 0x4e1d5ba is 26 bytes inside a block of size 48 free'd
==66825==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A582: main (manual_list.cpp:86)
==66825==  Block was alloc'd at
==66825==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==66825==    by 0x10A508: main (manual_list.cpp:85)
==66825==
Ghost
==66825==
==66825== HEAP SUMMARY:
==66825==     in use at exit: 0 bytes in 0 blocks
==66825==   total heap usage: 5 allocs, 5 frees, 74,896 bytes allocated
==66825==
==66825== All heap blocks were freed -- no leaks are possible
==66825==
==66825== For lists of detected and suppressed errors, rerun with: -s
==66825== ERROR SUMMARY: 12 errors from 5 contexts (suppressed: 0 from 0)
maryannleshaj@DESKTOP-HMH6I4P:~/EECE2140/Lab8_Memory$
```

This is the use after free bug